



The protective effects and mechanism of Ruyi Zhenbao Pill, a Tibetan medicinal compound, in a rat model of osteoarthritis

Qien Li ^a, Jing-Yi Xu ^c, Xin Hu ^c, Jun Li ^c, Xian-Ju Huang ^{c,d,*}, Zhou-Yang Wu ^{b,**}, Da-Gui Wang ^c, Yue-Bin Ge ^c

^a Tibetan Medical College, Qinghai University, Xining, PR China

^b Department of Anesthesiology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, PR China

^c College of Pharmaceutical Science, South-Central Minzu University, Wuhan, PR China

^d Ethnopharmacology Level 3 Laboratory, National Administration of Traditional Chinese Medicine, South-Central Minzu University, Wuhan, 430074, PR China

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ABSTRACT

Ethnopharmacological relevance: Ruyi Zhenbao Pill (RZP) is a prescribed Tibetan formulation for the treatment of white-pulse-disease, yellow-water-disease as well as pain-related disease. RZP is composed of 30 medicinal materials including herbal medicine, animal medicine and mineral medicine. They are widely used in the Tibetan area to treat cerebrovascular disease, hemiplegia, rheumatism, and pain diseases for centuries.

Aim of the study: The aim of the present study was to evaluate the anti-osteoarthritis function of RZP and to clarify the underlying mechanisms.

Materials and methods: The active components in RZP were identified using HPLC methods. Osteoarthritis (OA) animal model was established via intra-articular injection of papain in rat knees. After the administration of RZP (0.45, 0.9 g/kg) for 28 days, the clinical observation was conducted, and pathological changes as well as serum biochemical indexes were detected. Moreover, therapeutic targets and pathways of RZP were discussed.

Results: The results showed that RZP could suppress knee joint swelling and arthralgia, thus relieving joint pain and inflammation in OA rats. Microcomputed tomography (μ CT)-based physiological imaging and staining pictures confirmed the therapeutic effects of RZP on OA symptoms including knee joint swelling and structural changes with progressive inflammation in OA rats. RZP could promote the synthesis or inhibit the degradation of COLII, attenuate OA-induced OPN up-regulation and thus relieve the OA symptom. Furthermore, RZP (0.45–0.9 g/kg) could all ameliorate the imbalance of biomarkers related to OA such as MMP1, TNF- α , COX2, IL-1 β and iNOS in knee joints or serum.

Conclusion: In conclusion, RZP could effectively relieve inflammatory reaction induced by OA injury and the formulation could be applied to the treatment of OA therapy.

1. Introduction

Osteoarthritis (OA) is one of the most common forms of arthritis in clinic. General clinical symptoms such as joint pain, difficulty walking, and disability have a main impact on health-related quality of life (Hao et al., 2021). OA is not only a traditional articular cartilage disease but also a whole-joint disease including synovial tissue and synovial fluid (Roos and Arden, 2016). Increasing evidence suggests that in synovial fluid from OA patients, synovial lining cells and chondrocytes can produce and/or respond to various pro-inflammatory cytokines and

reactive oxygen species (ROS) (Kosinska et al., 2013; Sirše, 2022), bringing about the intra-articular stress response and tissue damage, leading to inflammatory/oxidative stress in chondrocytes and resulting in progressive joint degeneration, which have always expressed as hallmarks of OA (Ahmed et al., 2016).

The current treatment options remain controversial. The management of OA is unsatisfactory for a significant number of patients who neither respond to the existing conservative treatment nor achieve indications of joint replacement surgery. Accumulating evidence suggests that the inflammatory microenvironment in synovial fluid in the initiation of OA may be mainly due to early synovial inflammatory damage

* Corresponding author. South-Central Minzu University, 182 Minyuan Road, Wuhan, 430074, PR China.

** Corresponding author. Huazhong University of Science and Technology, Wuhan, PR China.

E-mail addresses: xianju@mail.scuec.edu.cn (X.-J. Huang), wzy6638@sina.com (Z.-Y. Wu).

Abbreviations

ALP	alkaline phosphatase
ALT	Alanine aminotransferase
ANOVA	one-way analysis of variance
BCA	bicinchoninic acid
BMD	bone mineral density
BS	bone surface
BS/TV	bone surface density
BV	bone volume
BV/TV	bone volume fraction
CaHA	calcium hydroxyapatite
CC	calcified cartilage
CMC	Sodium carboxymethylcellulose
COX2	Cyclooxygenases 2
CSG	chondroitin sulfate & glucosamine treatment
HA	hyaluronic acid
HC	hyaline cartilage
H&E	hematoxylin and eosin
HPLC	high performance liquid chromatography
IL-1 β	Interleukin-1 beta
iNOS	inducible nitric oxide synthase
MDA	malondialdehyde

MMP1	Matrix metalloproteinase 1
NO	Nitric oxide
NSAIDs	non-steroidal anti-inflammatory drugs
OA	Osteoarthritis
OPN	Osteopontin
OTC	over the counter
PVDF	polyvinylidene fluoride
RNS	reactive nitrogen species
ROI	region of interest
ROS	reactive oxygen species
RZP	Ruyi Zhenbao Pill
SMI	structure model index
SOD	Superoxide Dismutase
Tb.N	trabecular number
Tb.PF	trabecular pattern factor
Tb.Sp	trabecular spacing
Tb.Th	trabecular thickness
TC	total cholesterol
TG	triglyceride
TNF- α	Tumor Necrosis Factor-alpha
TPT	Thermal pain threshold
μ CT	Microcomputed tomography

(Lin et al., 2022; Zhang et al., 2022.). The main goal of the treatments for knee OA is to act at an early stage to halt these degenerative and inflammatory aspects and improve mobility and function. Most of the current strategies include non-steroidal anti-inflammatory drugs (NSAIDs), cyclooxygenase 2 (COX2) inhibitors and intra-articular treatments with anti-inflammatory agents, viscosupplements and biologic agents, which provide only partial and temporary pain relief and entail several limitations and risks (Sa et al., 2020; Gambari et al., 2022). Due to the side effects and limited action of existing drugs for OA, more effective therapeutic medicine needs to be discovered for preventing OA progression.

The discovery of new natural compounds is of great significance for drug exploitation and development in the treatment of OA. Supportive evidence has been offered that Chinese herbal medicine, either adjuvant therapy or monotherapy, reduces the visual analog score, Western Ontario and McMaster Universities Osteoarthritis Index, and Lequesne index and improves the Lysholm score and overall effective rate in patients with knee osteoarthritis (Xu et al., 2022; Lin et al., 2022.). Small molecules of herbal origin (Zhang et al., 2022) and some dietary supplements (Gugliandolo et al., 2020; Sirše, 2022) have also been reported to help manage joint diseases by regulating various signaling pathways related to inflammation and oxidative stress.

Tibetan medicine, a traditional medical system in Chinese medicine, has exhibited prominent effects in clinical practice for thousands of years. OA is classified as Yellow-water disease and described in *Four*

Medical Records, one of the Tibetan classic work (Yutuo, 1983). Compound Ruyi Zhenbao Pill (RZP), of which the Tibetan name is རུའི་འཇམ་ལོ་མོ་ལྷ་མོ་, is also known as Sangpei Nuobu in Chinese Pinyin and was first recorded in the Tibetan Medical Literature *Collection of Secrets* (Wang-dui, 2014). It is a traditional formulation for the remedy of white-vein-disease and yellow-water-disease according to the Tibetan medicine theory. It has a clinical medication history of more than 1300 years. In traditional Tibetan clinical medication, after the prescription is prepared into pills, it is taken with boiled water or yellow rice wine at noon or midnight to treat chronic diseases, gout, arthritis, leprosy, stiffness diseases, lumbar muscle strain, congestion, blood heat and chronic fever caused by the invasion of epidemic heat and turbulent heat. With the development of the pharmaceutical industry, the production process of RZP has been continuously standardized, and the quality standard has been improved gradually. It has been approved as a provincial hospital preparation and is widely used in clinical practice. Since 2002, it was approved by the Chinese National Medical Products Administration (national medicine permission number: Z6300289) and has been officially sold in the country, mainly used for the treatment of common and frequently occurring diseases such as white-pulse-disease (nervous system disease), pain and arthritis.

It consists of 30 medicinal materials, including herbal, animal and mineral medicines, which have been checked with “The Plant List” (www.theplantlist.org) and “The world Flora Online” (<http://www.worldfloraonline.org/>), as shown in [Supplementary Table 1](#).

It possesses much traditional efficacy such as relieving fever, restoring consciousness, opening orifices, and relaxing muscles and joints (Wang et al., 2021). Most recent investigations focus on its therapeutic effect on white-vein-disease, i.e., cerebral ischemic disease. However, it also exhibits protective effects against inflammatory diseases such as arthritis (a yellow-water-disease in Tibetan medicine). Tibetan doctors consider that the mechanism of white-vein-disease is consistent with some modern diseases, such as nerve pain, sequelae of stroke, rheum arthritis and cervical spondylopathy.

This study was thus conducted to evaluate the anti-arthritis effect of RZP and to disclose its mechanism. The protective effect of RZP against papain-induced OA rats was evaluated, and the mechanism of action was further discussed using molecular biological methods. The results will clarify the clinical application of traditional Tibetan compounds.

Table 1
HPLC conditions.

Subject	Column temperature (°C)	Flow rate (mL/min)	UV detection (nm)	Mobile phase(A: acetonitrile-B: 0.1% phosphoric acid solution)
Ellagic acid	30	1.0	254	16%: 84%
Eutannin	30	0.8	280	18%: 82%
Chromatographic fingerprints	30	1.0	210, 280	0–26 min, 5–13% A; 26–50 min, 13–24% A; 50–85 min, 24–35% A; 85–110 min, 35–67% A

2. Methods

2.1. Preparation and analysis of RZP

RZP was provided by Jinhe Tibetan Medicine Co., Ltd. According to the original constituents of this formula, multiple components of RZP were characterized by high-performance liquid chromatography (HPLC) performed on an Agilent 1200 system (Agilent Technologies, USA) using a Thermo Fisher C18 column (250 mm × 4.6 mm, 5 μm). The appropriate HPLC conditions were applied to different ingredients as listed in Table 1.

2.2. Animal models and experimental design

SPF Wistar rats (weight 240 ± 20 g, 20 male and 20 female) were purchased from Huazhong Agriculture University in Wuhan, P.R. China (NO. 42000600040790). The rats were housed under standard circumstance according to the guidance of the Laboratory Animal Center and the Laboratory Animal Ethics Committee of South-Central Minzu University (No. SCXK-E-2021-0089). All the animals were permitted five days to acclimate to the environment before starting the experiment.

According to the Drug Standard of the Ministry of Public Health (Edit, 1995) in China, the clinical dosage of RZP is 4–5 g. Based on the average human body mass of 70 kg, we administered an intermediate dose of 0.071 g kg⁻¹ to humans. The dosage for rats was calculated to be 6.3 times the human dose, and thus the intermediate dose for the rats in the RZP group was 0.45 g kg⁻¹. Then, 0.45 g kg⁻¹ was preferred as the low-dose group and 0.9 g kg⁻¹ was designed as the high-dose group in the experiment. The drug solution was created by adding 0.5% sodium carboxymethylcellulose (CMC) to make a suspension. The detailed schematic protocol of the OA model is shown in Fig. 2A as previously described (Hao et al., 2021). On the first day of the experiment, the rats were administered 0.15 ml of 4% papain solution. This injection site covered the intermalleolar fossa to the outer edge of the inferior patellar tendon. The same treatment was carried out on days 4 and 7. On day 8, induced OA rats were separated into four groups (n = 7 for each group) at random as follows.

- (a). Model group: OA rats with normal saline;
- (b). RZP group at low-dose: OA rats administered 0.45 g/kg RZP via oral gavage;
- (c). RZP group at high-dose: OA rats administered 0.9 g/kg RZP via oral gavage;
- (d). Positive control group: OA rats administered chondroitin sulfate & glucosamine treatment via oral gavage, referred to as “CSG”, 125.0 mg/kg;
- (e). Control group: normal rats only treated with saline.

Drug was administered daily in the morning on the 8th day and lasted for 4 weeks. The rats were observed twice and weighed everyday. Blood samples were collected after the last administration and the serum was stored at -80 °C. Then the rats were put to death and the knee joints of the affected side were separated to conduct microcomputed tomography (μCT). Osteoarticular cartilage was collected for histological detection. Organs including liver, lung, heart, kidney and spleen were weighed the organ indices were calculated respectively. The related tissue was frozen at -80 °C for further detection.

2.3. Pain behavior and joint swelling assessment

Prior to drug treatment (7 d), pain behaviors were assessed as a baseline and then once a week until 28 d post-papain injection. The thermal pain threshold (TPT) was measured using a hot plate (Yixing Instrument, Co. Ltd, RCY-2) maintained at 55 ± 0.5 °C, and the time of licking the hind paws or attempting to jump out from the beaker was detected as the latency period (Ren et al., 2012). A set of calibrated von

Frey filaments (0.4–26 g; North Coast Medical, China) was used to determine mechanical hyperalgesia in the hind paws (Huang et al., 2021).

From day 1 and 4 weeks following papain treatment, knee diameters (mean of 3 readings per knee) were measured every 4 days. The joint swelling was assessed by the D-value of diameter (mm) between the ipsilateral and contralateral knees (Huang et al., 2021). Swelling rate of knee joint was calculated using the percentage of the D-value in paw diameter relative to the control group by a formula $[(\text{sample D-value} - \text{control D-value}) / \text{control D-value}] \times 100\%$.

2.4. Osteoarticular cartilage histopathology

Osteoarticular cartilage from different rats was fixed with 10% neutral-buffered formalin for later histopathological examination. The sections with a thickness of 4 μm were stained with hematoxylin and eosin (H&E) (Huang et al., 2021; Yu et al., 2016). Morphological alterations were observed under the optical microscope (Bresser 54-03001). A modified Mankin histologic scoring system was employed to evaluate cartilage damage (Fusco et al., 2020; Cordaro et al., 2019), from 0 (normal histology) to 12 (complete disorganization and hypocellularity).

2.5. Immunohistochemistry detection

Immunohistochemistry detection was employed to analyze the expression of collagen II (COLII) and osteopontin (OPN) protein in the osteoarticular cartilage of rats. Firstly, osteoarticular cartilage slices were fixed in 3% H₂O₂ and 3% normal goat serum. Then they were incubated with Collagen II (1:200, Santa Cruz Biotechnology, USA) and OPN rabbit polyclonal antibodies (1:100, Santa Cruz Biotechnology, USA) in 0.01 mol/L phosphate-buffered saline solution at 4 °C overnight. Then, the sections were washed and incubated with secondary antibodies (goat anti-rabbit HRP, 1:200, Servicebio) at normal temperature for 50 min. Each slice was stained with DAB (Servicebio, Wuhan, China), and the coloration time was controlled under a microscope. Then, the nucleus was stained by hematoxylin. After the slices were dehydrated by alcohol and dimethylbenzene, photos were taken under a microscope.

2.6. Microcomputed tomography (μCT)

Specimens were scanned using a Bruker Micro-CT Skyscan 1276 system (Kontich, Belgium). The scan conditions were as follows: voxel size 10 μm, medium resolution, 85 kV, 200 mA, 1 mm Al filter and integration time 384 ms. Measurements of density were calibrated to the manufacturer's calcium hydroxyapatite (CaHA) phantom. Data analysis was operated by the manufacturer's testing software. Reconstruction was completed by NRecon (Ver 1.7.4.2). 3D images were obtained from contoured 2D images by methods according to the distance changes of the grayscale original images (CTvox; version 3.3.0). 3D and 2D analyses were conducted via CT Analyser software (Ver 1.18.8.0). Analyses of the bone microarchitecture were performed in a region of interest (ROI). Bone morphometry of the cortical and epiphyseal trabecular bone of the tibia and femur was carried out as previously described (Besler et al., 2021). Measurements of bone volume fraction (BV/TV, [%]), bone volume (BV, [mm³]), bone surface (BS, [mm²]), bone surface density (BS/TV, [mm⁻¹]), trabecular number (Tb. N, [mm⁻¹]), trabecular thickness, (Tb. Th, [mm]), trabecular spacing (Tb. Sp, [mm]), trabecular pattern factor (Tb. PF, [mm⁻¹]), structure model index (SMI) and bone mineral density (BMD, [g.cm⁻³]) were determined.

2.7. Oxidative stress and inflammatory indices measurements in serum

The oxidative stress and inflammatory indices including malondialdehyde (MDA), superoxide dismutase (SOD), matrix metalloproteinase 1

Table 2
Primers used for qRT-PCR.

Gene	Primer
β -actin-F	CTTCTGGGCATGGAGTCCT
β -actin-R	GGAGCAATGATCTTGATCTT
TNF α -F	GTGTCCCAACATTCATATTGTCAGT
TNF α -R	TGGGAAGAGAAACCAGGGAGA
iNOS-F	TGAGTTCCGAAGCAAGCCAA
iNOS-R	AGACCTCAACAGAGCCCTCA
COX2-F	TCTCAACCTCTCTACTACTAC
COX2-R	GCACGTAGTCTTCGATCACT
IL-1 β -F	TGGGATAGGGCTCTCTTGC
IL-1 β -R	CCATGGAATCCGTGCTTCTCT

(MMP1) and nitric oxide (NO) in serum were detected by commercial kits (Jiancheng Bioengineering Institute, Nanjing, China).

2.8. Real-time RT-PCR

Four types of pro-inflammatory cytokines in the serum, i.e. TNF- α , IL-1 β , Cyclooxygenases 2 (COX2) and inducible nitric oxide synthase (iNOS) were detected by real-time RT-PCR, thus further interpret RZP-induced regulation in the intermediate process of OA. In brief, total RNA of knee joints was harvested using TRIzol Reagent and mRNA was amplified using a One Step SYBR Green Prime Script™ RT-PCR kit (Takara, Japan). The cycling procedure was as follows: step 1, 95 °C for 30 s, 95 °C for 5 s; step 2, 40 cycles at 58 °C for 34 s. The relative magnitude of gene mRNA was determined by the $2^{-\Delta\Delta Ct}$ data analysis method and normalized to β -actin. Primers used in the study are listed in Table 2.

2.9. Western blot analysis

The knee joints were pulverized and homogenized in ice-cold lysis buffer combined with a protease inhibitor cocktail (Servicebio, China) using an automated tissue homogenization instrument (Servicebio, China). Then supernatant proteins were obtained through centrifugation of tissue lysates at 4 °C. They were quantified by bicinchoninic acid (BCA) protein concentration assay kits (Servicebio, G2026, China). IL-1 β , TNF- α , COX2 and iNOS were detected respectively with β -actin (Servicebio, China) as an internal control. Same amounts of protein were detached by SDS-polyacrylamide gel electrophoresis and later transferred onto a polyvinylidene fluoride (PVDF) membrane blocked with 5% skimmed milk powder. Afterwards, the membrane was incubated at 4 °C overnight with corresponding primary antibodies (1:1000)

followed by probes with secondary antibodies (1:10000). Visualization of protein was accomplished using chemiluminescence reagents on Alpha and Adobe imaging system (Alpha Innotech, alphaEaseFC; Adobe, Adobe Photoshop).

2.10. Statistical analysis

The statistical analysis was conducted by SPSS software (Ver. 18.0). All data were presented as means $\pm$ SD. Multiple comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's test. A probability (*P*) value of less than 0.05 was considered statistically significant.

3. Results

3.1. Analysis of main ingredients in RZP

As shown in Fig. 1, six major constituents of RZP were identified as chebulagic acid (peak 1, 36.46 min), ellagic acid (peak 2, 37.54 min), eutannin (peak 3, 42.01 min), quercetin (peak 4, 57.34 min), kaempferol (peak 5, 68.47 min) and isosalantolactone (peak 6, 107.30 min). The contents of compounds in RZP were approximately 2.02 mg/g (ellagic acid) and 5.25 mg/g (eutannin) based on the calculations.

Among the six chemicals, ellagic acid and eutannin, which are abundant compounds in RZP and rich in the fruit and seed of *Terminalia chebula* Retz., exhibit significant cytoprotective, anti-arthritis and anti-proliferative activities (Bag et al., 2013).

3.2. Animal experiment

3.2.1. Changes in body weight, joint swelling and pain threshold

No mental states or abnormal behaviors were displayed in the control group and no rats died during the experiment. No obvious sexual difference observed among all the groups during the clinical study. There were no obvious differences in organ indices of the liver, lung, heart, kidney and spleen among all the groups (Table 3). On day 7 after the first administration of papain, obvious decreases in body weight, lameness, weak pedaling, swollen knee joint and higher sensitivity to mechanical or hot stimulation were shown in OA rats (Fig. 2). However, the body weight returned to normal gradually (Fig. 2B) and there was little difference among all the groups at the end of the experiment. RZP (0.45–0.9 g/kg) and CSG treatments could both significantly suppress knee joint swelling from day 10 (see Fig. 2C) and ameliorate hot and mechanical allodynia in OA rats (see Fig. 2D and E), suggesting that RZP

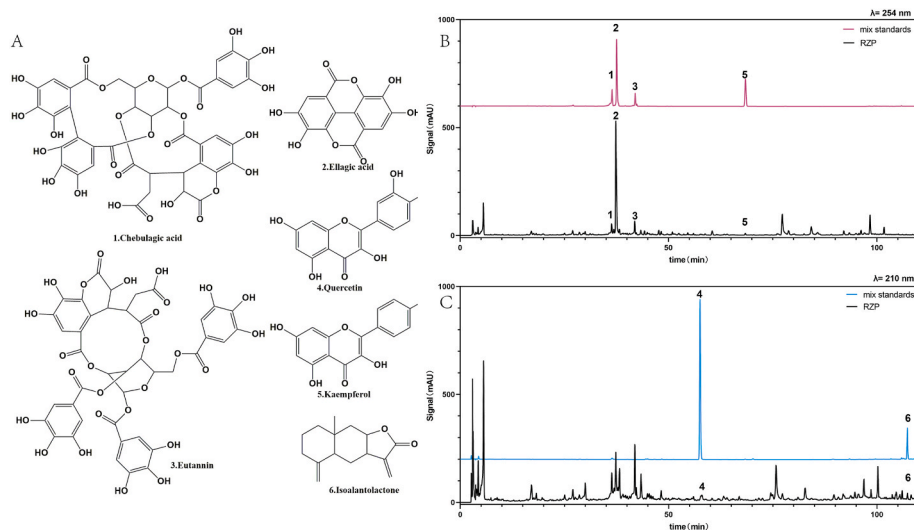


Fig. 1. The chemical structures and the typical HPLC chromatographic profile of RZP. (A) The chemical structures of chebulagic acid, ellagic acid, eutannin, quercetin, kaempferol and isosalantolactone; (B) HPLC chromatographic profile of RZP and mixed standards at 254 nm; (C) HPLC chromatographic profile of RZP and mixed standards at 210 nm. Chebulagic acid (peak 1, 36.46 min), ellagic acid (peak 2, 37.54 min), eutannin (peak 3, 42.01 min), quercetin (peak 4, 57.34 min), kaempferol (peak 5, 68.47 min) and isosalantolactone (peak 6, 107.30 min).

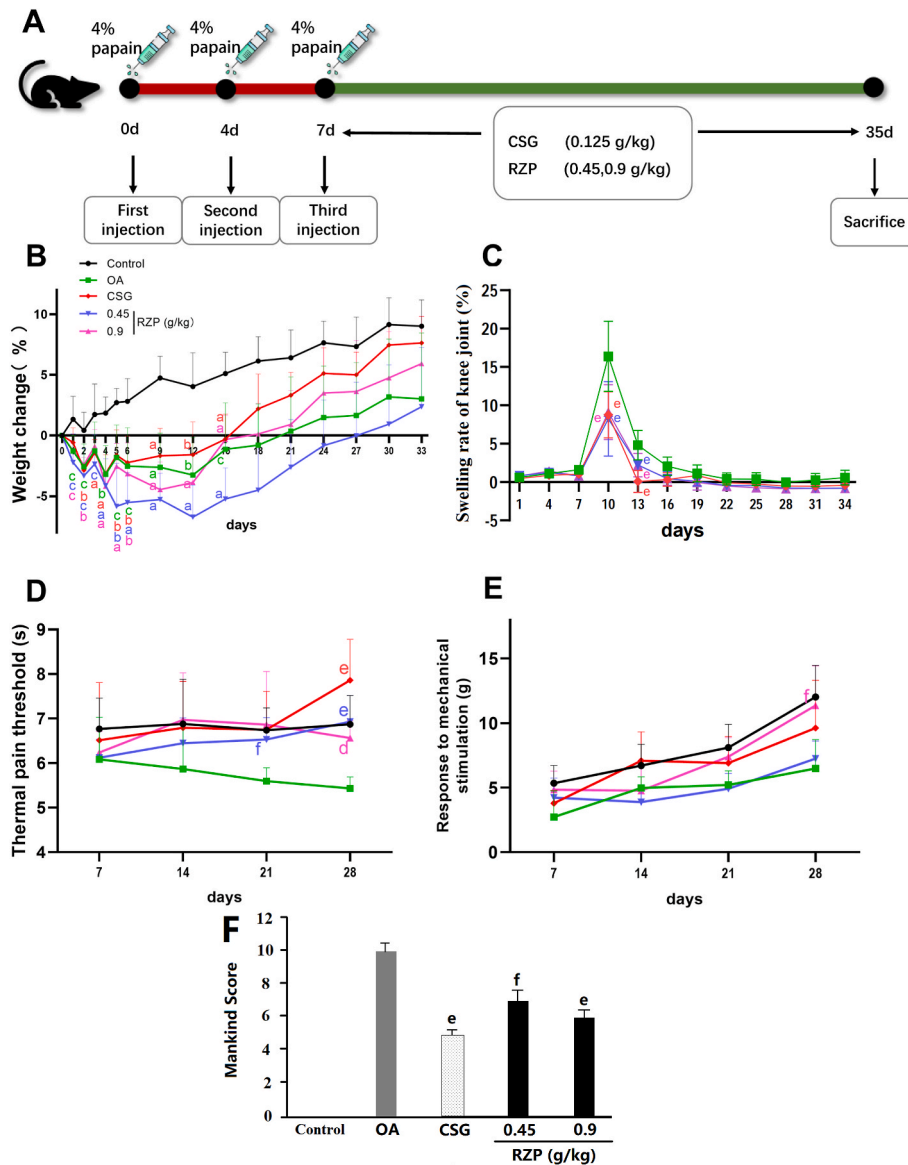


Fig. 2. Experimental design and clinical observation of RZP on the influence of OA in rats. (A) Experimental design; (B) weight change; (C) swelling rate of knee joint; (D) thermal pain threshold; (E) mechanical pain threshold; (F) Mankin scores of pathological changes; Data are means $\pm$ SD. a $P < 0.001$, b $P < 0.01$, c $P < 0.05$, compared with the control group; d $P < 0.001$, e $P < 0.01$, f $P < 0.05$, compared with the OA group.

Table 3

The organ indexes of heart, liver, spleen, lung and kidney.

Group	Heart (mg/g)	Liver(mg/g)	Spleen (mg/g)	Lung(mg/g)	Kidney (mg/g)
Control	3.014 $\pm$ 0.205	26.693 $\pm$ 1.796	2.228 $\pm$ 0.214	4.531 $\pm$ 0.303	6.159 $\pm$ 0.166
OA	3.006 $\pm$ 0.224	24.855 $\pm$ 1.645	2.055 $\pm$ 0.337	4.579 $\pm$ 0.565	6.203 $\pm$ 0.391
CSG	2.915 $\pm$ 0.237	26.339 $\pm$ 1.003	2.158 $\pm$ 0.127	4.512 $\pm$ 0.893	6.219 $\pm$ 0.313
RZP-0.45 g/kg	3.311 $\pm$ 0.405	25.084 $\pm$ 1.654	2.166 $\pm$ 0.167	4.262 $\pm$ 0.316	6.220 $\pm$ 0.744
RZP-0.9 g/kg	3.185 $\pm$ 0.192	25.073 $\pm$ 1.916	2.144 $\pm$ 0.123	4.175 $\pm$ 0.288	6.097 $\pm$ 0.497

Note: N = 7, mean $\pm$ SD.

could suppress knee joint hyperalgesia and swelling, therefore generating a therapeutic effect on arthritic inflammation in OA rats.

3.3. Micro-CT (μ CT) analysis of RZP on the knee joint of OA rats

The right knee joint in OA group rats displayed redness and swelling (Fig. 3A). The scan result of μ CT and 3D remould are displayed in Fig. 3. A remarkable multihole structure occurred in the knee-joints of OA rats, indicating bone loss and corrosion on the surface of the knee joints. CSG and a high dose of RZP attenuated the above injury to some extent, and their knee joints became smoother (Fig. 3B).

The tomographic image is shown in Fig. 3C. The Tb. N was decreased, Cg. Th was thinner, and the cavity areas of bone marrow were larger in the OA samples than in the control samples, indicating significant bone loss and bone rarefaction (shown by the arrow in Fig. 3C). Moreover, Fig. 3D shows that the thickness of cartilage in the thigh bone and thin bone turned thinner in the OA group compared with the normal rats. CSG and RZP ameliorated the above injury to some extent.

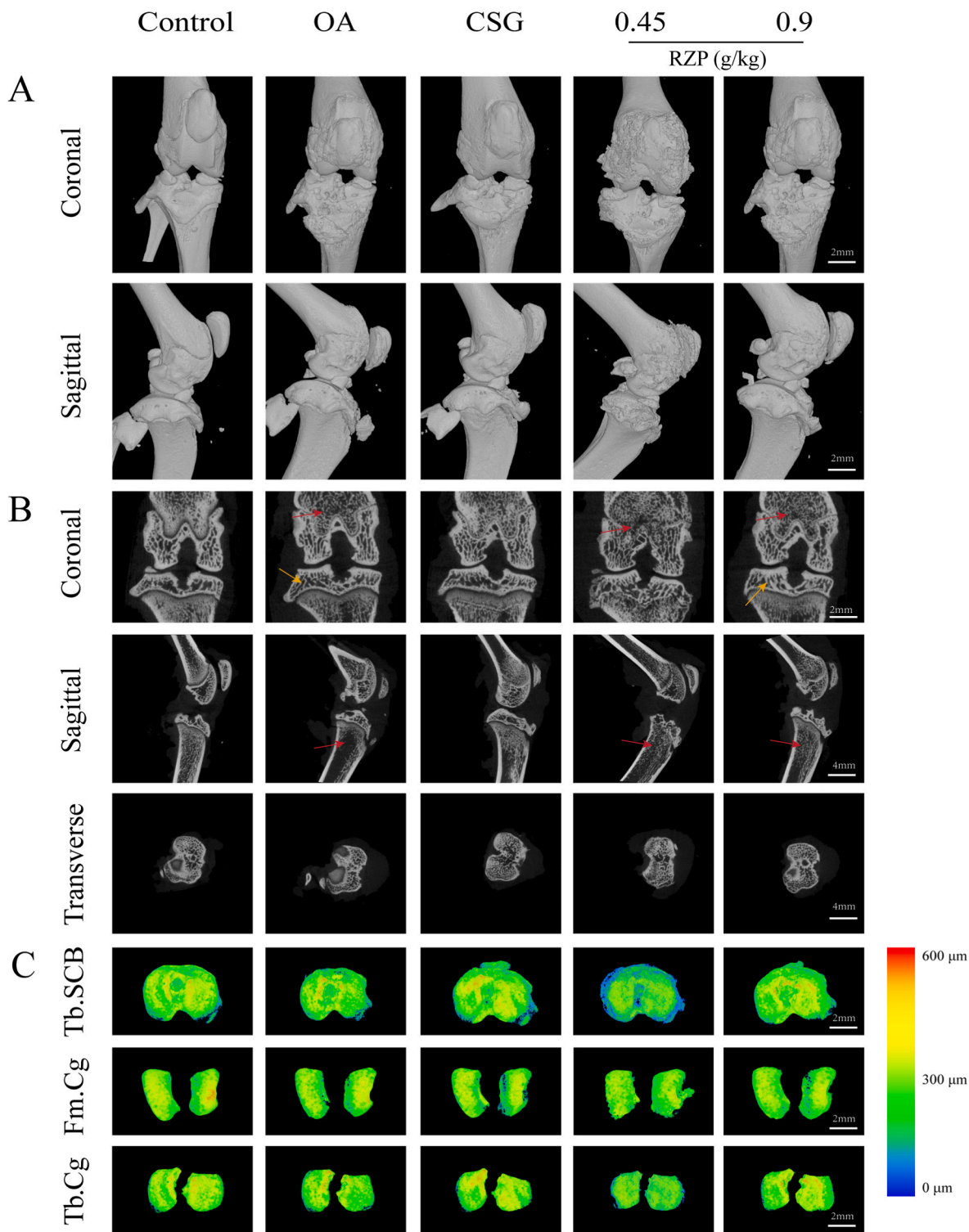


Fig. 3. Arthritis of rat joint in each group. (A) appearance of the knee joint; (B) 3D reconstruction pictures of bone shape; (C) micro-CT 3-view diagrams; (D) thickness maps of femoral and tibial cartilage compartments. Reproducibility of the 3D QMA in terms of intraclass correlation coefficient (ICC) and precision errors (PE) expressed in absolute and a percentage of the coefficient of variation of the repeated measure.

The relative quantitative measures are listed in Table 4. Comparison between the OA and control joints demonstrated that BV/TV, BS/TV, Tb. N and BMD were notably decreased in the OA animals, indicating bone loss in OA joints. Similarly, Tb. PF, Tb. Sp and SMI were remarkably increased in the OA joints, suggesting the trend of trabecular bone changing from plate shape to rod shape. Although there was little difference in Tb. Th between the OA model and normal joints, the Tb. Th in

the RZP group was higher than that in the normal control group, suggesting the positive effect of RZP on the anabolism of bone. After 4 weeks of administration, CSG and RZP could both attenuate the above parameters, suggesting that RZP could promote the composition of trabecular bone and increase bone mass, thus preventing osteoporosis.

Table 4

Relative parameters of rat knee joint in micro-CT scanning.

Indexes	Control	OA	CSG	RZP-0.45 g/kg	RZP-0.9 g/kg
BV/TV (mm ³)	49.17 ± 5.24	36.95 ± 4.57 [#]	41.21 ± 5.04 [*]	38.01 ± 4.79	42.59 ± 5.22 [*]
BS/TV (mm ³)	13.21 ± 3.01	10.96 ± 2.12 [#]	11.22 ± 2.33 [*]	10.73 ± 1.87	11.66 ± 2.41 [*]
Tb.Th(mm)	0.16 ± 0.03	0.13 ± 0.02	0.14 ± 0.02	0.14 ± 0.03	0.15 ± 0.04
Tb.N (mm ⁻¹)	3.58 ± 0.42	2.73 ± 0.33 [#]	3.43 ± 0.45 [*]	3.08 ± 0.41 [*]	3.51 ± 0.48 [*]
Tb.Sp(mm)	0.25 ± 0.03	0.31 ± 0.04 [#]	0.26 ± 0.04 [*]	0.26 ± 0.04 [*]	0.27 ± 0.04 [*]
Tb.PF (mm ⁻¹)	0.94 ± 0.10	5.73 ± 0.68 [#]	3.59 ± 0.51 [*]	4.37 ± 0.62 [*]	4.01 ± 0.53 [*]
SMI	0.22 ± 0.03	1.21 ± 0.23 [#]	0.72 ± 0.09 [*]	0.81 ± 0.19 [*]	0.78 ± 0.10 [*]
BMD (g*mm ⁻³)	0.71 ± 0.09	0.46 ± 0.07 [#]	0.61 ± 0.08 [*]	0.62 ± 0.07 [*]	0.63 ± 0.07 [*]

Note: [#]*P* < 0.05, OA vs Control, ^{*}*P* < 0.05, treated vs OA, *N* = 3, mean ± SD; Tb.SBP=Tibial subchondral bone, Fm.Cg = Femoral cartilage, Tb.Cg = Tibial cartilage.

3.4. Influence of RZP on the pathological changes in the knee-joint of OA rats

H&E staining results suggested that significant pathological changes could be observed in the knee joint of OA rats (Fig. 4A). The OA rats

displayed severe tissue erosion and cartilage cell disarrangement on the surface of the cartilage. In addition, the infiltration of inflammatory cells, part of the extracellular matrix totally degraded and the cartilage four-layer structure were unclear in OA rats, as shown by arrows. CSG significantly alleviated the infiltration of inflammatory cells, extracellular matrix degradation and cartilage four-layer structure unclearance. RZP (0.45 g/kg)-treated rats showed rough surfaces and cracks on the cartilage, and the infiltration of inflammatory cells was still obvious. However, the four-layer structure was clear. RZP (0.9 g/kg)-treated rats showed smooth cartilage surfaces and cell arrangements, the cartilage matrix was complete, and the four-layer structure was clear, indicating a dose-dependent ameliorative effect of RZP on OA rats, as evidenced by the Mankin scores (Fig. 2F).

The proteoglycan of articular cartilage assessed by safranin O staining is presented in red, and the bone tissue assessed by solid green is presented in blue-green (Fig. 4B). The osteocyte number and distribution of proteoglycans in the OA group were decreased and could be attenuated by CSG and RZP (0.9 g/kg).

As shown in Fig. 4C, hyaline cartilage (HC) was stained dark blue, and calcified cartilage (CC) was stained blue by toluidine. The tidal line between HC and CC in the control group was complete, and cells in each layer were arranged neatly. However, tidal line interruption and CC layer thinning could be observed in OA rats, indicating repeated pathological changes in a tidal structure under high pressure. RZP and CSG treatment attenuated injury and led to CC thickening, osteocyte number increase and cell arrangement.

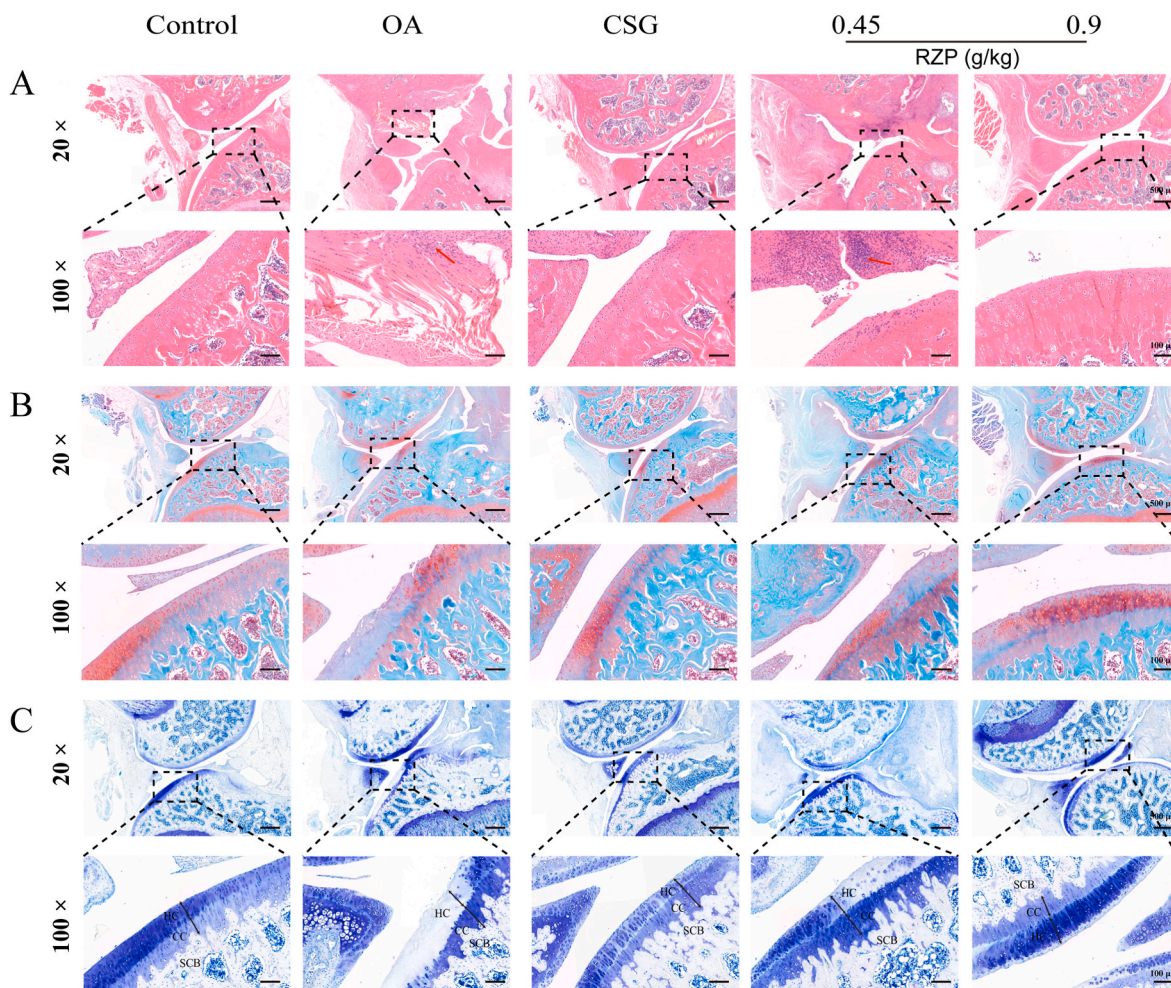


Fig. 4. Staining results of rat joint in each group. (A) HE staining; (B) Safranin-O-fast green; (C) Toluidine blue staining. Note: HC=Hyaline cartilage, CC=Calcified cartilage, SCB=Subchondral bone. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

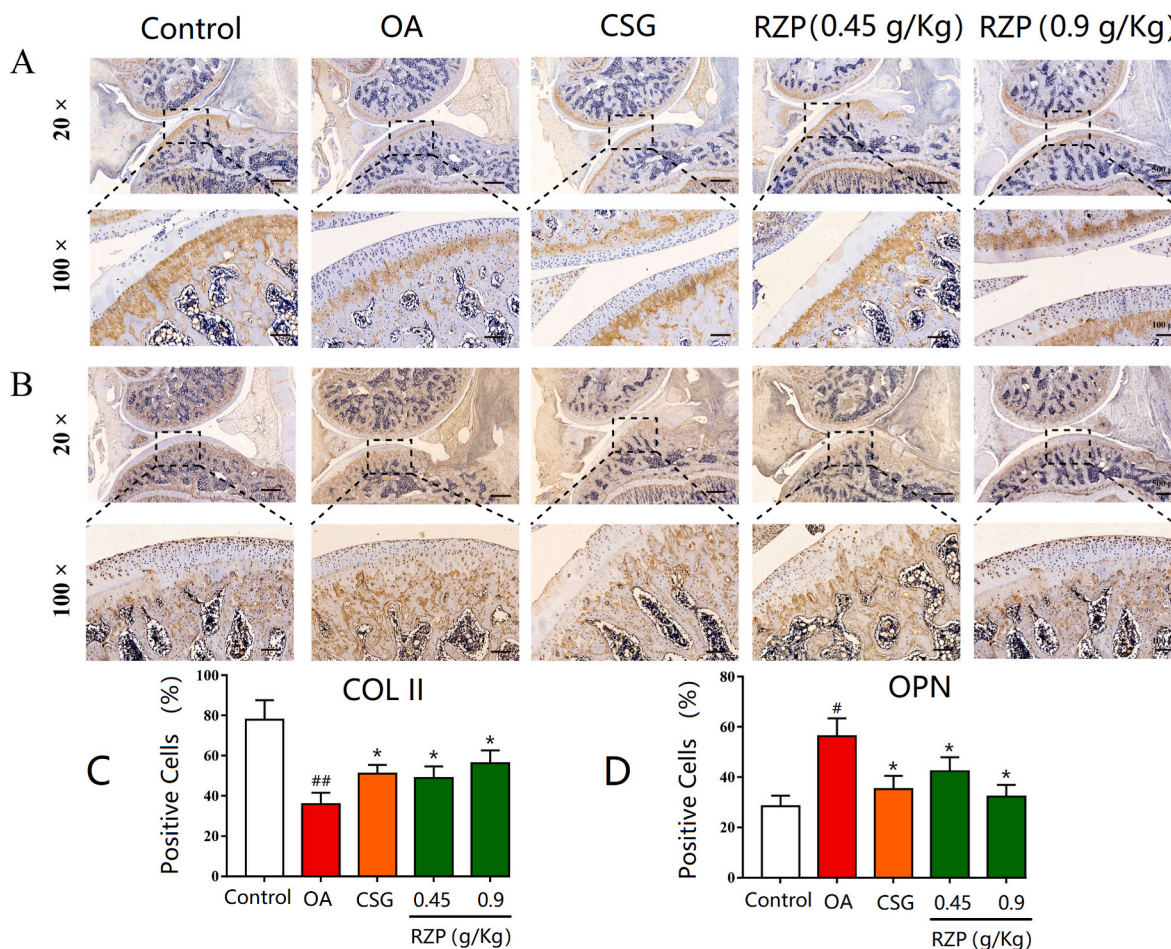


Fig. 5. Immunohistochemistry images of COL II and OPN immunoreactivity of rat joints in each group. A and C, Representative images of a COLII-stained plot with percentages of COLII-positive cells were quantified. B and D, Representative images of a OPN-stained plot with percentages of OPN-positive cells were quantified. $n = 3$, mean $\pm$ SD. $^{##}P < 0.01$, $^{\#}P < 0.05$, vs Control group, $^{**}P < 0.01$, $^{*}P < 0.05$, vs OA group.

3.5. Influence of RZP on COLII and OPN expression in the knee-joint of OA rats

COLII and OPN were quantitated using immunohistochemistry. As shown in Fig. 5, positive expression of COLII and OPN proteins in the knee-joint of OA rats of each group appeared to be yellowish-brown. COL II is mostly expressed in the intercellular matrix of the calcified cartilage layer, and OPN is often expressed in articular cartilage. OPN was significantly increased (Fig. 5 B), and COLII had a down-regulation trend (Fig. 5 A) in the OA group compared to the control group. CSG and RZP treatments increased COLII expression to varying degrees, and the color deepened. On the other hand, CSG and RZP administration attenuated the increase in OPN in the knee joint of OA rats. These findings implied that RZP could promote the synthesis or inhibit the degradation of COL II and attenuate OA-induced OPN up-regulation, thus relieving OA symptoms.

3.6. Influence of RZP on oxidative stress indices and proinflammatory cytokines production in serum

As shown in Fig. 6, increases in the serum concentrations of MMP1 and NO, down-regulation of SOD and up-regulation of MDA were displayed OA group, indicating the inflammation and oxidative stress that occurred in the model. RZP (0.45–0.9 g/kg) attenuated the tendency in a dose-dependent manner.

3.7. The influence of RZP on OA-induced iNOS, COX2, TNF- α and IL-1 β mRNA levels and protein expression

qRT-PCR results (see Fig. 7A) showed that iNOS, COX2, TNF- α and IL-1 β mRNA were all up-regulated in the knee joints of OA rats. CSG and RZP (0.45–0.9 g/kg) ameliorated the imbalance to some extent. The Western blot results are shown in Fig. 7B. Similar to the CSG group, RZP (0.45–0.9 g/kg) attenuated the up-regulation of TNF- α , iNOS, COX2 and IL-1 β protein in OA rats, suggesting that RZP could effectively relieve the inflammatory reaction induced by OA injury.

4. Discussion

Traditional Tibetan medicine has been applied to cure OA-related diseases for long time. Identification of the active constituents and clarifying basic pharmacological mechanisms is still a great challenge for the subject of Tibetan medicine. RZP is a traditional Tibetan prescription for OA therapy. Due to the multiple components and targets, its mechanism is still unclear to us. The present study revealed that RZP partially restored OA-induced changes by ameliorating symptoms (reducing inflammation, minimising pain, and maintaining joint function), attenuating the hydrolysis of collagen, inhibiting inflammatory cytokines and oxidative stress, suggesting that the anti-osteoarthritis effects of RZP may be due to down-regulation of the expression/production of pro-inflammatory cytokines. However, because of individual differences, two-dose (0.45 and 0.9 g/kg) of RZP showed no discernible difference in body weight, swelling rate of knee joint and oxidative

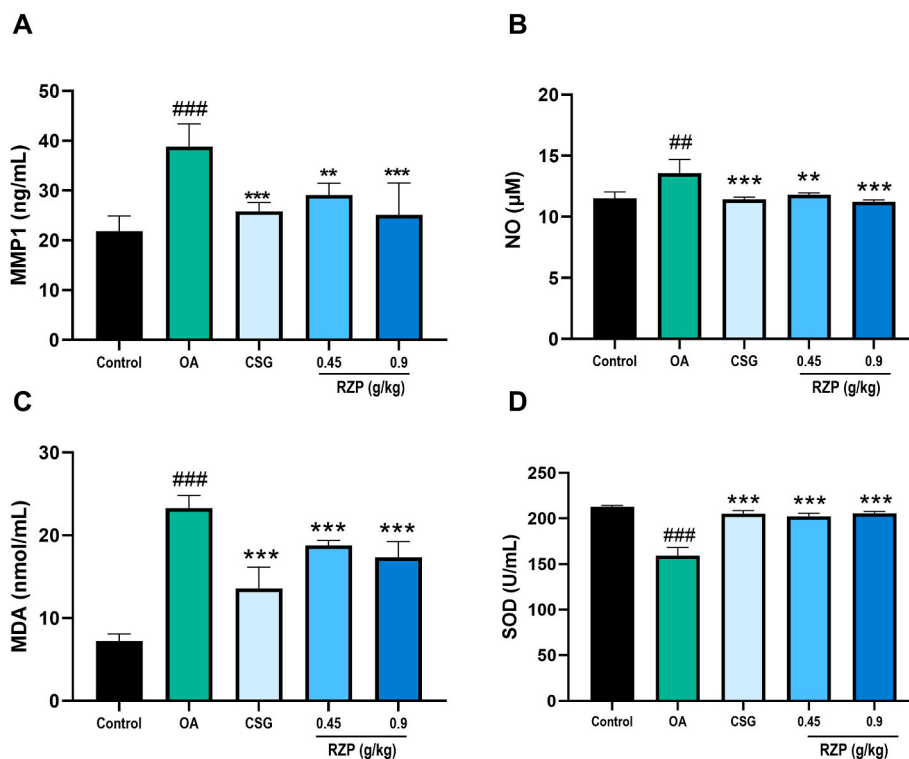


Fig. 6. Influence of RZP on MMP1 and NO concentrations and oxidative stress indices in OA rats. N = 5, mean $\pm$ SD. ^{###} $P < 0.001$, ^{##} $P < 0.01$, vs. the control group, ^{***} $P < 0.001$, ^{**} $P < 0.01$, vs. the OA group.

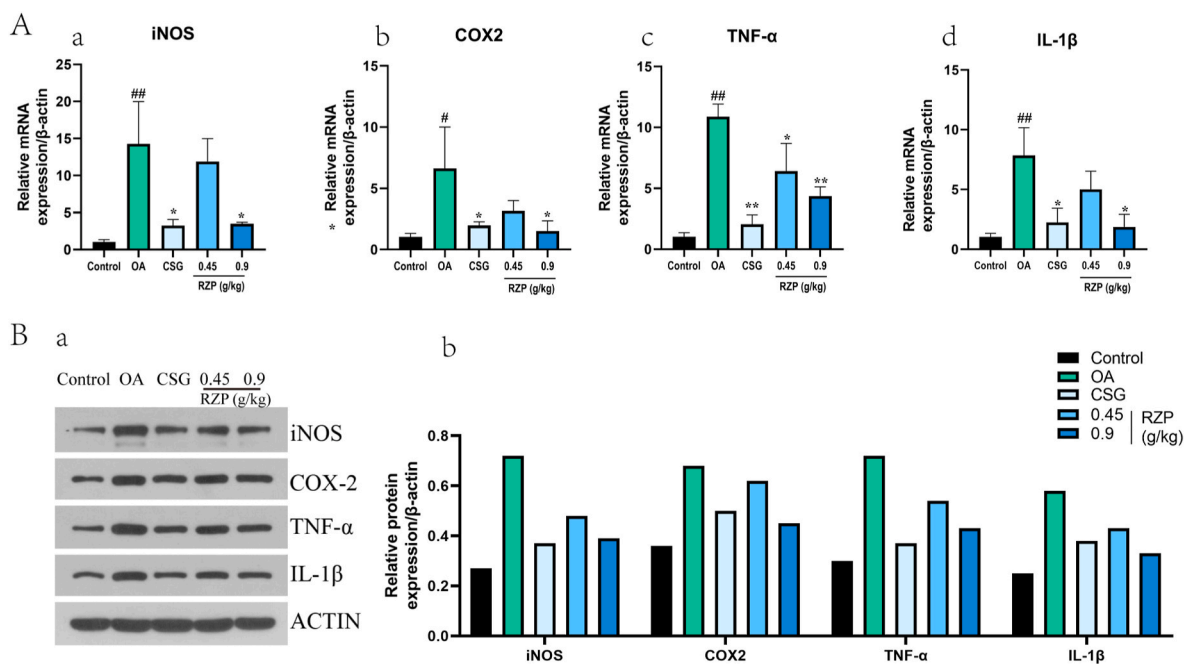


Fig. 7. The effect of RZP on OA-induced iNOS, COX2, TNF-α and IL-1β expression. (A) effects of RZP on the iNOS, COX2, TNF-α and IL-1β mRNA levels in the joints of OA rats; (B) effects of RZP on the iNOS, COX2, TNF-α and IL-1β protein levels in the joints of OA rats. n = 3, mean $\pm$ SD. ^{##} $P < 0.01$, [#] $P < 0.05$, vs Control group, ^{**} $P < 0.01$, ^{*} $P < 0.05$, vs OA group.

stresses in OA rats, but has obvious differences in the indexes such as pathological changes and inflammatory indices. More acute animal models and more extensive doses of RZP should be adopted in our future study.

Joint inflammation plays a central role in the pathological process of OA and, together with other pathological processes, ultimately leads to a

disruption of joint homeostasis and biomechanics and causes joint pain and stiffness (Ju et al., 2020; Hao et al., 2021; Gambari et al., 2022). Anti-inflammation remains one of the primary treatments for OA therapy (Zhao et al., 2022; Villanueva Romero et al., 2018; Britti et al., 2017). There is also a close association between OA progression and oxidative stress. Inflammation is a key factor associated with cartilage

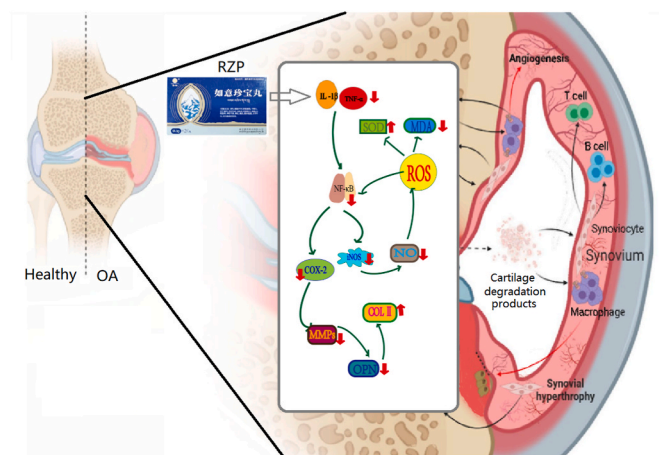


Fig. 8. The potential mechanism of the anti-OA action of RZP.

matrix degradation, and the presence of inflammatory cytokines may cause the loss of articular cartilage homeostasis via metabolic changes and markedly accelerate joint damage. High levels of pro-inflammatory cytokines such as IL-1 β , IL-6, IL-10, and TNF- α have been found in several experimental animal models of cartilage degradation and the synovial fluid of OA patients (Di Paola et al., 2016). For instance, COX2 is the key mediator of inflammatory symptoms including increased body temperature and pain (Kavitha and Velraj, 2017). TNF- α , IL-1 β , and IL-6 are mainly related to immunomodulation and the inflammatory reaction in the body (Goldring and Otero, 2011). TNF- α promotes other cytokines and plays an important role in cartilage damage. IL-1 β causes apoptosis in chondrocytes by relying on endogenous NO (Blanco et al., 1995). MMPs are a superfamily of proteases that mainly contribute to the extracellular matrix degradation and tissue remodeling. MMP-1, 3 and 13 are believed to play key roles and are expressed together with IL-1 β in the development of OA, causing degradation of the cartilage matrix (Wang et al., 2011).

OPN, an extracellular matrix glycoprotein abundant in bone in the progress of metabolic homeostasis of normal articular cartilage, has been considered a potential inflammatory cytokine. It can be secreted by many cell types including chondrocytes, osteoblasts, osteocytes and osteoclasts (Gao et al., 2013). OPN also plays a significant role in regulating proteoglycan, hyaluronic acid (HA), and COLII (Liu et al., 2020). It is publicly agreed that pro-inflammatory mediators such as TNF- α and IL-1 β , can promote MMPs production through the activation of mitogen-activated protein kinases (MAPKs) and NF- κ B (Huang et al., 2021). The present findings conceived that RZP inhibits the production of MMP-1, OPN and elevated the content of COLII via inhibition of pro-inflammatory cytokines and the inflammatory/oxidative pathway in OA (Fig. 8).

The increase of inflammatory factors could promote cartilage proteolysis, which partially illustrates the persistent reduction in the knee joint (Chen et al., 2017). Administration with RZP significantly relieved knee joint swelling, arthralgia, and joint dysfunction, inhibited the inflammatory factors and oxidative indices and regulated MMP imbalance. The anti-inflammatory mechanism of RZP on OA rats maybe through the suppression of the IL-1 β /NF- κ B pathway activation, as shown in Fig. 8. TNF- α is a potent pro-inflammatory cytokine and, along with IL-1 β , acts as a catabolic factor for the cartilage. NF- κ B, another major pathway activated by TNF α /IL-1 β , is involved in almost all pathological processes of OA (Zhang et al., 2022; Lee et al., 2022). Other than inducing catabolic enzymes, the activation of NF- κ B induces the expression of major pro-inflammatory mediators of OA such as iNOS and COX-2, thus resulting in oxidative stress. Oxidative stress and an imbalance in the pro-oxidant/antioxidant status can result in reactive oxygen species (ROS) production, leading to the formation of MDA, and

consumption of SOD in OA cartilage and chondrocytes. As shown in Fig. 8, RZP treatment could suppress IL-1 β /TNF- α , thus inhibiting NF- κ B, leading to down-regulation of COX2 and iNOS. The inhibiting of iNOS could reduce oxidative stress by promoting SOD activity and decreasing MDA contents. On the other hand, the down-regulation of COX2 could inhibit the production of MMP-1, and OPN and elevated the content of COLII, indicating that RZP possesses a therapeutic effect in OA rats.

Overall, the results indicated that the inflammatory mediators including TNF- α , IL-1 β , and IL-6, and oxidative stress indices could be adopted to evaluate the therapeutic potential of anti-osteoarthritis medicine. Our study illustrated that RZP exhibited convincing therapeutic efficiency by inhibiting the inflammatory reactions through its multiple compounds via the suppression of the IL-1 β /NF- κ B pathway.

5. Conclusion

Taken together, RZP plays an anti-osteoarthritis role by inhibiting inflammatory and oxidative reactions to relieve joint pain and protect articular cartilage, indicating the effectiveness of this traditional formula towards the remedy of OA. The present investigation brings a deep knowledge of the pharmacological efficacy of RZP, which may lead to further successful utilization of RZP for clinical therapy.

Author's contribution

Xian-Ju Huang, Zhou-Yang Wu and Qi-En Li designed the investigation. Xian-Ju Huang supervised the project. Jing-Yi Xu, Da-Gui Wang and Yue-Bin Ge performed the experiments. Xin Hu, Jun Li and Zhou-Yang Wu analyzed the data. Xian-Ju Huang, Qi-En Li and Jun Li wrote the manuscript. All authors have read and approved the final manuscript prior to submission.

CRedit authorship contribution statement

Qien Li: Project administration, Writing – review & editing. **Jing-Yi Xu:** Visualization, Writing – original draft. **Xin Hu:** Resources, Funding acquisition. **Jun Li:** Formal analysis, Software. **Xian-Ju Huang:** Supervision, Funding acquisition, Writing – review & editing. **Zhou-Yang Wu:** Project administration, Conceptualization. **Da-Gui Wang:** Visualization, Data curation. **Yue-Bin Ge:** Validation, Formal analysis, All authors have read and approved the final manuscript prior to submission.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116255>.

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